

Name: _____

Skill: knowledge

Mitosis

Cut out the descriptions and diagrams for the five stages of mitosis and arrange them in the correct order on page 2 of this worksheet.

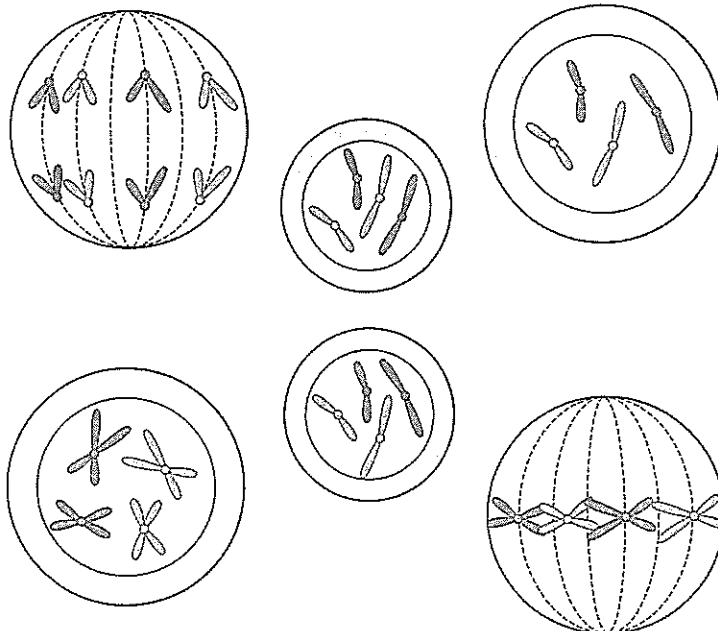
Chromosomes separate and move to the ends of the cell.

Two pairs of chromosomes are visible.

Chromosomes are doubled but attached at a point called the centromere.

Membranes form to produce two daughter cells.

Chromosomes line up along the 'equator' of the cell.



Meiosis

Cut out the descriptions and diagrams for the seven stages of meiosis and arrange them in the correct order on page 3 of this worksheet.

Chromosomes are doubled but attached at a point called the centromere.

Chromosomes separate and move to the ends of each cell.

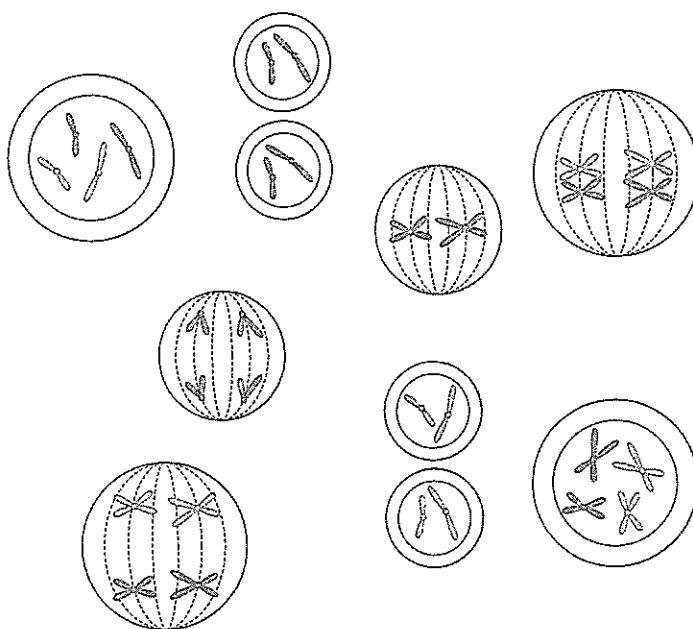
Membranes form to produce two daughter cells.

Chromosomes line up along the 'equator' of the cell.

Homologous chromosomes line up along the 'equator' of the cell.

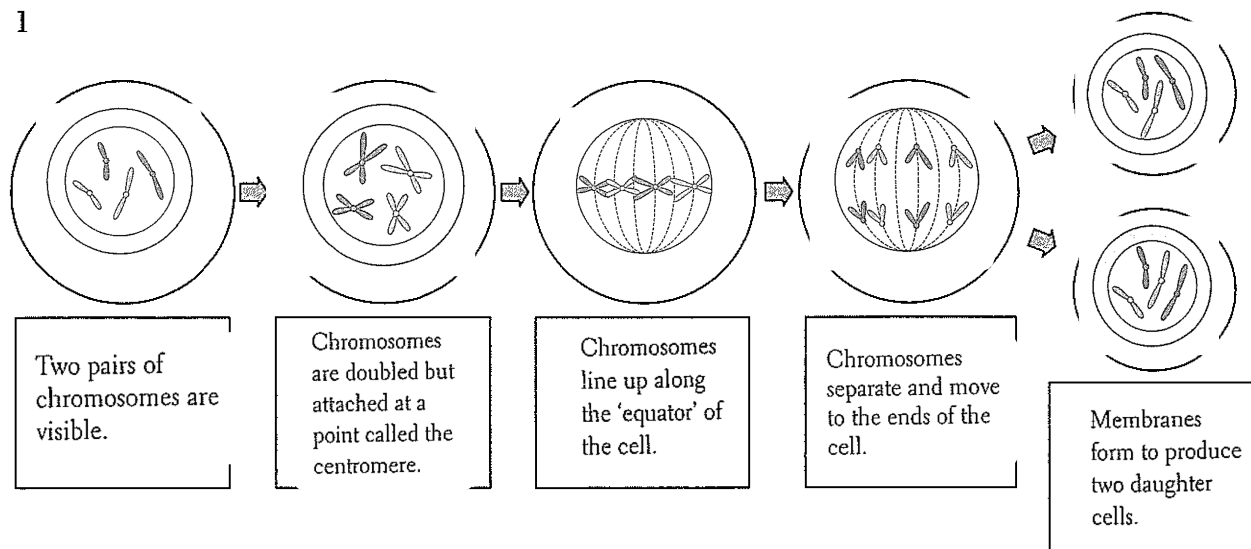
One of each pair of chromosomes moves to the ends of the cell.

Two pairs of chromosomes are visible.



Mitosis

1



2 Define the following terms:

a homologous pair

Two chromosomes with the same set of genetic information.

b diploid

Normal number of chromosomes.

c haploid

Half the normal number of chromosomes.

d daughter cell

Produced from the copying and/or splitting of a cell.

3 Identify the parts of the body in which mitosis occurs.

All body cells except for the reproductive cells (ova or sperm).

4 Outline the purpose of mitosis in:

a multicellular organisms

For growth and replacing damaged cells.

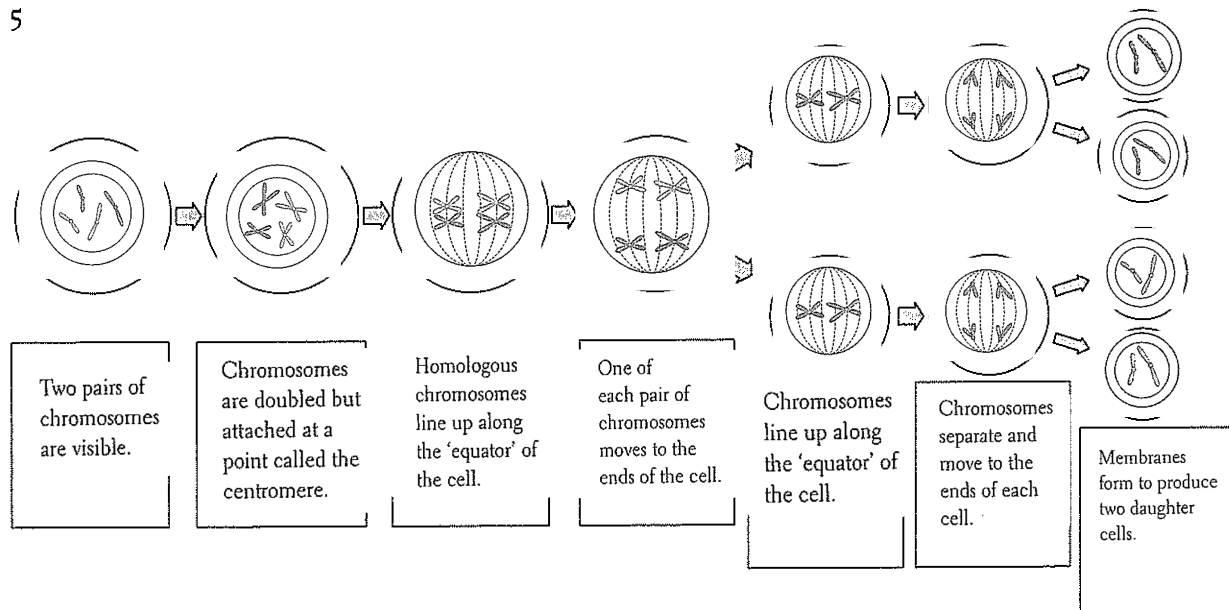
b single-celled organisms.

Asexually reproducing to give a new individual.

Name: _____

Meiosis

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6 Identify the parts of the body in which meiosis occurs.

Reproductive cells - ova or sperm.

7 Outline the purpose of meiosis.

Produce reproductive cells with half the normal number (haploid number) of chromosomes.

8 What names are given to the daughter cells produced during meiosis:

a in males? Sperm

b in females? Ova

9 Complete the following table to summarise the differences between mitosis and meiosis.

	Mitosis	Meiosis
Number of daughter cells produced	2	4
Number of chromosomes in parent cells of humans	46	46
Number of chromosomes in daughter cells in humans	46	23